

A Coupling-Diagnostic Spectral Screen for Damping-Margin Planning in Inverter-Dominated Industrial Power Grids

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Abstract—This poster proposes a damping-margin planning diagnostic for inverter-dominated industrial power grids. The central result is a rational graph-filter correction that detects when classical Laplacian/frequency screens can recommend line or control actions that improve modal stiffness while reducing the true small-signal damping margin.

Index Terms—inverter-based resources, small-signal stability, damping ratio, industrial power systems, quadratic eigenvalue problem, graph signal processing, line reinforcement, planning diagnostics

PROBLEM AND IAS RELEVANCE

Industrial and utility power systems are increasingly operated with inverter-based resources (IBRs), grid-following controls, grid-forming controls, and heterogeneous damping mechanisms. In this regime, a classical network-strengthening action can improve a graph-frequency metric while reducing the true small-signal damping margin. This creates a practical planning problem for industry-facing grids: how can an engineer screen line reinforcements, damping additions, or synchronous-generator-to-IBR transitions without solving a full quadratic eigenvalue problem (QEP) for every candidate action?

CLASSICAL BASELINE AND SOTA GAP

Starting from the linearized model

$$M\ddot{\theta} + D\dot{\theta} + L\theta = p, \quad (1)$$

classical screening diagonalizes the inertial Laplacian $\tilde{L} = M^{-1/2}LM^{-1/2}$, forms modal stiffnesses ν_k , and estimates damping from $\delta_k = q_k^T \tilde{D}q_k$, where $\tilde{D} = M^{-1/2}DM^{-1/2}$. This is exact only when damping and stiffness commute. Full/reduced QEP eigensolves and modal perturbation can capture non-proportional damping, but they are heavier and less interpretable as fast planning screens.

PROPOSED CONTRIBUTION

The proposed contribution is to expose the missing non-proportional damping term through a second-order correction to the critical pole:

$$\Delta s_c = \frac{s_0^2}{2s_0 + \delta_c} \sum_{\ell \neq c} \frac{E_{c\ell}E_{\ell c}}{s_0^2 + \delta_\ell s_0 + \nu_\ell}, \quad (2)$$

$E = \text{offdiag}(Q^T \tilde{D}Q).$

This expression is interpreted as a rational graph filter of the operator pair $(\tilde{L}, \tilde{D})$ evaluated at the critical pole. It therefore explains why a polynomial graph filter of $\tilde{L}$ alone cannot represent IBR damping: the relevant information is the off-diagonal modal coupling induced by non-proportional damping.

PRELIMINARY RESULTS

Preliminary results support the mechanism. In a reproducible six-bus synthetic case with moderate non-proportional damping, the frequency-only top line reinforcement raises the critical modal stiffness by +0.0931 under a +10% line-weight change, but the full-QEP damping margin drops from 0.04204 to 0.04068. The second-order correction predicts the harmful direction with 16.8% relative error on the damping-margin movement, while a damping-aware line choice improves the margin. On an IEEE39-derived reduced surrogate, 35/46 line reinforcements raise the critical modal stiffness while lowering $\zeta_{\min}$. Estimator tests on the same surrogate reduce median relative error from 0.468% for the diagonal screen to 0.0321% for the second-order correction, and a reduced grid-following inverter model with PLL memory improves from 2.16% to 0.09% median error over 399 stable cases.

POSTER MESSAGE

The intended IAS contribution is an interpretable planning diagnostic, not a replacement for full small-signal analysis. The workflow is: use the diagonal graph screen when modal coupling is weak, apply the rational correction when off-diagonal damping matters, and fall back to reduced or full QEP solves when modes are clustered or strongly coupled. This directly supports IAS-relevant planning questions in industrial and converter-dominated power systems: which reinforcement actually improves damping, where damping should be added, and when a cheap spectral metric is unsafe.